

Project Documentation

1. Introduction

Project Title

OptiCrop - Smart Agricultural Production Optimization Engine

Team Members

Team Member	Role
D.Goutham	Team Member

Mentor: No Mentor Assigned

2. Project Overview

Purpose

OptiCrop is a machine learning-based agricultural recommendation system designed to optimize crop production using environmental and soil parameters. It integrates ML algorithms with a Flask web application to provide intelligent crop recommendations based on key environmental factors, helping farmers, agricultural researchers, agribusiness companies, and policymakers make data-driven decisions that improve productivity, sustainability, and resource utilization.

Features

- Smart crop recommendation based on soil nutrients (N, P, K) and environmental parameters
- Crop suitability and environmental assessment for a given set of conditions
- Support for agricultural research and policy planning through data-driven pattern analysis
- Comparison of multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means) to select the best-performing model
- Interactive, responsive web interface for entering parameters and viewing instant predictions
- Trained model persistence using Pickle (.pkl) for fast reuse during prediction